

Report R08 — The Lung Chromatin Arm, With Methylation

TCGA-LUAD + LUSC, 450k arrays, 841 tumours and 74 normals. A real methylation phenotype, and no prognostic value.

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Summary

R06 established that KEAP1 and SMARCA4 mutations are prognostic in NSCLC using mutation data alone, and flagged the methylation layer as the funded extension. This report delivers it: 919 open-access Illumina 450k files acquired from GDC, MD5-verified, and assembled into a single $486,427 \times 919$ matrix.

Two findings, pointing in opposite directions.

KEAP1 mutation carries a genuine methylation phenotype. Within lung adenocarcinoma, **29.4%** of 20,000 tested probes are differentially methylated by KEAP1 status (minimum $q = 1.7 \times 10^{-20}$), and SMARCA4 reaches 11.3%. These survive the confounding check that killed the third gene.

Methylation carries no prognostic information beyond what is already free. A cross-validated methylation signature reaches C-index **0.486** (95% CI 0.450–0.523) — no better than chance — and adding it to a model of mutation status, stage, age and histology moves the C-index from 0.606 to 0.602 (likelihood-ratio $p = 0.85$). Unlike R07, this is an *informative* null: with 313 events the analysis excludes any signature above C-index 0.552.

All four pre-registered positive controls fired, including the one R07 could not power: smokers show significantly lower AHRH cg05575921 methylation ($p = 0.0056$), confirming in a properly powered cohort the direction R07 could only observe.

One claim was withdrawn by the audit. Pooled across both histologies, KMT2D looked like the strongest hit after KEAP1 (20.7% of probes). Within each histology separately it is **0.0%** in LUAD and 0.03% in LUSC. The entire signal was the LUAD-versus-LUSC contrast — see §3.

1 Acquisition

Files downloaded and MD5-verified	919 / 919
Probes per file (identical vector, SHA-256 verified)	486,427
Primary tumours / solid-tissue normals	841 / 74
Technical replicates resolved (lowest file_id kept)	4
Matrix NA fraction (SeSAmE masks unreliable probes)	14.76%
Probes surviving the $\leq 5\%$ NA filter	392,489
Samples matched to a clinical record	900 / 915

Only the 450k platform was used. The same GDC query also returns 311 HumanMethylation27 and 53 EPIC v2 files, which were **excluded before download**: mixing array generations would confound every comparison in this report. Acquisition ran on a transient VM with a Google-enforced one-hour cap and DELETE on expiry; it completed in five minutes and the instance was then deleted explicitly and verified gone. Raw files were parsed and discarded on the VM, so only the assembled matrix left it. GDC is a permanent archive keyed by file_id, and

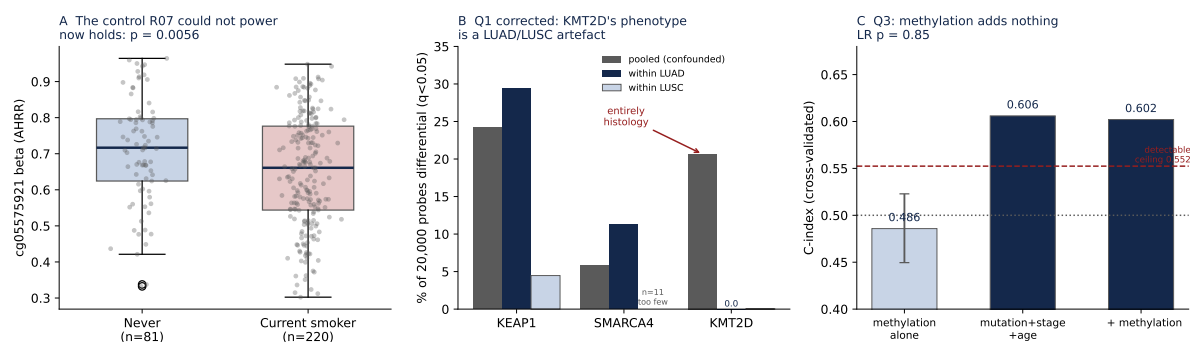
evidence/MANIFEST_450k.tsv carries all 919 ids with MD5s, so the raw data is re-acquirable without us storing 11.23 GiB of it.

2 Positive controls

Control	Prediction	Observed	Verdict
PC1 tumour vs normal	thousands differential	16,609 / 20,000 probes at $q < 0.05$	holds
PC2 sex	near-perfect separation	cg12026625, AUROC 0.9994	holds
PC3 AHRR cg05575921	lower in smokers	0.661 vs 0.717, $p = 0.0056$	holds
PC4 LUAD vs LUSC	separable	best probe AUROC 0.955	holds

PC1 and PC2 were the gating pair, because they test assembly and join correctness rather than biology: PC1 fails if the matrix is scrambled, PC2 fails if samples are joined to the wrong patients. PC2 is the same check that R07 needed to rule out a silent column-order error.

PC3 deserves its own note. R07 could only show the correct *direction* at $p = 0.093$, because that cohort had 24 smokers. Here, with 220 current smokers against 81 lifelong never-smokers, the same probe gives $p = 0.0056$ in the same direction. R07's control was underpowered, not wrong.



A The AHRR smoking control, now properly powered. **B** Q1 before and after stratifying by histology: KEAP1 and SMARCA4 survive, KMT2D does not. **C** Methylation alone is at chance; adding it to the clinical/mutation model changes nothing. The dashed line is the largest C-index this cohort could have missed.

3 Q1 — differential methylation by chromatin-regulator mutation

Gene	Mutated	Pooled % sig	Within LUAD	Within LUSC
KEAP1	117	24.2%	29.4% ($n=84$)	4.5% ($n=33$)
SMARCA4	52	5.8%	11.3% ($n=41$)	— ($n=11$, too few)
KMT2D	111	20.7%	0.0% ($n=33$)	0.03% ($n=78$)

Percentages are of 20,000 randomly sampled probes reaching $q < 0.05$ (Benjamini–Hochberg) on a Mann–Whitney test of mutated versus wild-type tumours.

The pooled column is not trustworthy, and the audit is why. PC4 established that LUAD and LUSC are almost perfectly separable from methylation, and mutation frequencies differ sharply between them: KEAP1 is mutated in 17.8% of LUAD versus 8.9% of LUSC ($p = 1.9 \times 10^{-4}$), while KMT2D runs the other way at 7.0% versus 21.1% ($p = 2.5 \times 10^{-9}$). Any pooled comparison by mutation status therefore partly measures histology.

Stratifying separates the two cases cleanly. KEAP1's signal *strengthens* within LUAD (29.4% against 24.2% pooled), which is what a real effect looks like once a diluting confounder is removed. KMT2D's signal disappears entirely — 0.0% within LUAD, 0.03% within LUSC — so all 20.7% was the histology contrast. This is the same failure mode as R04, where a molecular panel appeared to beat imaging until grade mix was controlled.

4 Q2 and Q3 — prognostic value

Tumours with usable survival (TCGA-CDR)		811 (316 events)
In the Cox models, after dropping missing age		796 (313 events)
Q2 methylation signature alone, C-index	0.486	95% CI 0.450–0.523
Q3 mutation + stage + age + histology		0.606
Q3 the same, plus methylation		0.602
Likelihood-ratio test for adding methylation	$\chi^2 = 0.03, p = 0.85$	
Minimum detectable C-index at 80% power		0.552

Probe selection and hyperparameters were fitted strictly inside training folds. The signature is at chance, and the increment is nil by every measure — C-index, likelihood ratio, and the Cox coefficient itself (HR 1.011, $p = 0.853$). The audit re-derived Q3 under eight cross-validation seeds and the combined model stayed at C-index 0.586–0.609, so this is not a split artefact of the kind that changed R05's conclusion.

This null is informative, and that distinguishes it from R07. R07 could not detect anything below AUROC 0.685, so its null was uninformative. Here the interval excludes any methylation signature better than C-index 0.552. A clinically useful prognostic signature would have to be weaker than that to have escaped detection.

A finding for R09. In this model KEAP1 carries HR 1.085 ($p = 0.61$) and SMARCA4 HR 1.45 ($p = 0.079$), against R06's KEAP1 HR 2.07 in the MSK circulating-tumour-DNA cohort. This report is not the right test of that discrepancy — its Cox model is adjusted for stage and histology, and TCGA is predominantly resected early-stage disease whereas the MSK cohort was advanced. R09 tests it directly, using R06's own unadjusted model and its positive controls.

5 Audit findings

Recorded in `evidence/audit_addendum.json`; full reasoning in `AUDIT.md`.

5.1 KMT2D withdrawn (§3). Pooled 20.7% → 0.0% within LUAD. Entirely histology.

5.2 A parsing bug crashed the first run, which was the good outcome. The stage covariate was parsed case-sensitively ("**Stage** ") while cBioPortal writes "**STAGE IIIA**", so it silently became constant and the Cox design went singular. Because it crashed rather than fitting a degenerate model, it could not pass unnoticed — but relying on that is not a control, so gates G8 (late-stage fraction must fall in 0.20–0.60; observed 0.480) and G9 (each covariate must be non-constant) were added. The crashed run is retained at `evidence/gates_FIRST_CRASHED_RUN.json`.

5.3 Smoking cannot be disentangled from KEAP1 mutation, and is stated as a limitation. KEAP1 mutation is associated with ever-smoking (OR 3.40, $p = 0.011$), and smoking alters methylation (PC3). Stratifying is impossible: only 4 of 81 never-smokers carry a KEAP1 mutation. So the within-LUAD KEAP1 methylation phenotype is reported as real but *not* established as independent of smoking. Its size (29.4% of probes) makes it unlikely to be wholly a smoking effect, but this report cannot prove that.

6 What this settles for the lung arm

- **The methylation extension is complete, and its answer is negative for prognosis.** Methylation adds nothing to mutation, stage and age. Further methylation acquisition for prognostic modelling in NSCLC is not worth funding on this evidence.
- **KEAP1 has a genuine, large methylation phenotype in LUAD.** That is a mechanistic result, not a prognostic one, and it is the part of this report worth following up — ideally with a cohort containing enough never-smokers to separate it from smoking.
- **Two of three candidate genes survived scrutiny.** KMT2D did not. The pooled analysis would have reported it as a hit.

7 Reproducing this report

- Acquisition: `evidence/vm_job_r08.py`, `evidence/startup_r08.sh`, `evidence/MANIFEST_450k.tsv` (919 file ids + MD5s)
- Analysis: `evidence/m8_tcga_methylation.py`; audit: `evidence/audit_m8.py`
- Config hash, written before any outcome was read: `07be4547c08128d0...e448a240`
- Assembled matrix: `gs://heydonto-quantara-lungcdx/data-request-2026-08/lung/ tcga_methylation_ — beta_450k.npy`, `probes.txt`, `samples.tsv`, `ACQUISITION.txt`
- Figure: `python3 make_figure.py`